For Examiner's Use

4 **(a)** Define *electric potential* at a point.

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……

…… [2]

(b) Two charged solid spheres A and B are situated in a vacuum. Their centres are separated by a distance of 30.0 cm, as shown in Fig. 4.1. The diagram is not drawn to scale.

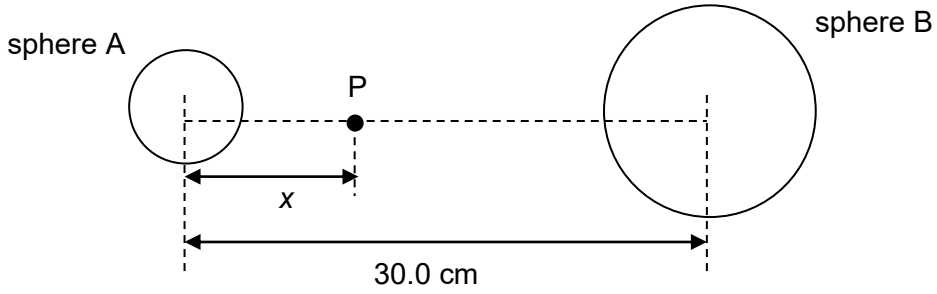

Fig. 4.1

The variation with distance x of the electric potential V_A and V_B due to sphere A and B independently is shown in Fig. 4.2.

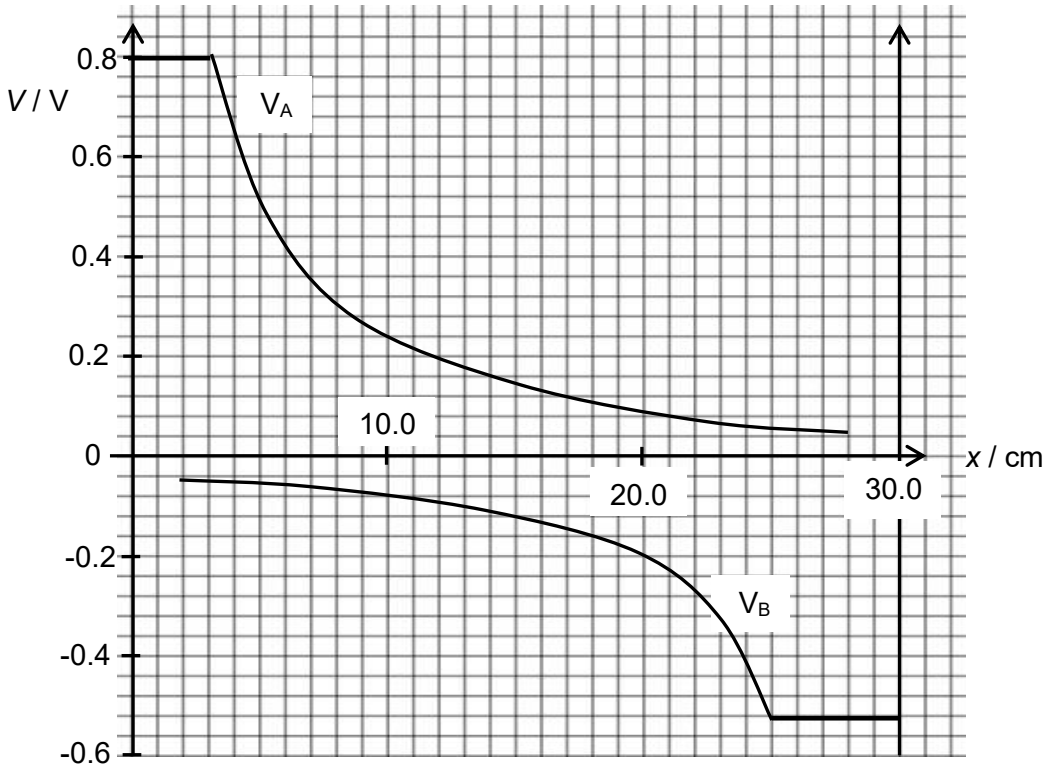

Fig. 4.2

(i) Using Fig. 4.2, state the radius of both spheres.

radius of sphere A = …… cm

radius of sphere B = …… cm [1]

(ii) State and explain the signs of both spheres.

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…….... [2]

(iii) Point P is at a distance x = 10.0 cm.
An alpha particle has kinetic energy E_K when at infinity.

Use Fig. 4.2 to determine the minimum value of E_K such that the alpha particle may travel from infinity to point P.

E_K = …… J [3]